Memorandum: AfyaPlus Generative AI Layer, Quality, Cost and Risk

To: CTO; Medical Director **From:** Olu, AI Engineering **Date:** 12 November 2026

Executive summary

We tested gpt-4o and gpt-4o-mini on 15 clinical questions across USSD, Mobile App and Web Portal. On a 1–5 clinical review scale gpt-4o averaged 4.8 and gpt-4o-mini 4.6; 6 of 6 model–channel combinations met every safety threshold. Sending each feature to the cheapest model that meets those thresholds moves monthly running cost (API fees plus clinician review) from $1,375.36 to $1,254.08, and drift monitoring first flagged a shift in patient message length in Month 2.

Quality performance

An independent AI reviewer (gpt-4o acting as judge) scored each answer against an approved clinical protocol. *Correctness* = clinically accurate; *groundedness* = makes no claims beyond the protocol, which is our guard against made-up advice ("hallucination"). Safety threshold: average ≥ 4/5 and no single answer below 3/5.

- **gpt-4o**: correctness 4.7/5 (±0.6 between answers), groundedness 4.7/5, relevance 5.0/5, helpfulness 4.9/5; weakest single answer scored 3/5 for correctness; wording overlap with the protocol (ROUGE-L, 0–1) 0.25. Meets safety thresholds for: Web Portal health FAQ, Mobile App medication help, USSD triage.
- **gpt-4o-mini**: correctness 4.4/5 (±0.7 between answers), groundedness 4.3/5, relevance 5.0/5, helpfulness 4.7/5; weakest single answer scored 3/5 for correctness; wording overlap with the protocol (ROUGE-L, 0–1) 0.25. Meets safety thresholds for: Web Portal health FAQ, Mobile App medication help, USSD triage.

Cost & efficiency

- 30-day API spend at today's split (75% gpt-4o-mini, 25% gpt-4o): $152.40 for 152,870 requests. gpt-4o handles 25% of requests but accounts for 85% of spend.
- Cost per request: gpt-4o-mini $0.00020, gpt-4o $0.00339 (17× more).
- Clinician review is the bigger cost: every model passed, so only a 2% spot audit is needed at $0.40 per reviewed answer (assumed ~2 minutes of clinical officer time). That still adds $1,222.96 per month, 8× the API bill.
- Routing by safety check: API fees $152.40 → $31.12; clinician review $1,222.96 → $1,222.96; net saving $121.28 (9%).

Operational risks

- **Drift** (a statistically significant change in incoming patient messages or answer quality compared with launch, measured on three months of simulated traffic): Month 2: answer quality (ROUGE-L) (-3% vs launch); Month 2: patient message length (+41% vs launch); Month 3: answer quality (ROUGE-L) (-18% vs launch); Month 3: response time (+32% vs launch); Month 3: patient message length (+66% vs launch).
- **Hallucination**: 0 of 15 gpt-4o-mini answers and 0 of 15 gpt-4o answers scored 2/5 or lower on groundedness; 0 reviewer outputs could not be read.
- **Budget**: day-30 spend is 98% of the $6.00 daily cap; on current growth, next month is projected at $169.95 (106% of the $160.00 monthly cap).

Recommended actions

1. **Route each channel by safety check**: Web Portal health FAQ → gpt-4o-mini; Mobile App medication help → gpt-4o-mini; USSD triage → gpt-4o-mini. Saves $121.28 per month, and every live model meets the clinical thresholds. Each channel was tested on only 5 questions, so move USSD triage gradually (start with 10% of triage traffic) and re-run the safety check weekly.
2. **Shorten the system prompt**: Moving from the chain of thought prompt to zero shot cuts API fees by $18.02 per month (58%). Safety condition: roll out only after the shorter prompt passes the same 15-question safety check.
3. **Re-test automatically when drift is detected**: when the alert log flags a change, automatically re-run the 15-question safety check (API cost $0.02 per run) and freeze routing changes until it passes, so declining quality is caught before it reaches patients.

Sources: evaluation/model_comparison.csv, quality_gate_summary.csv, llm_judge_matrix.csv; drift/drift_alert_log.json; cost/routing_savings.csv, cost_by_model.csv, cost_per_request.csv, prompt_config_savings.csv, budget_status.json. Pricing: OpenAI list prices in config.py. Data mode: live.